

Derivation for time evolution of mixed layer height

Tian Lang Liu

August 2026

Initial Profile

Begin with the initial buoyancy profile:

$$b_0(z) = \begin{cases} 0 & z < T \\ N_0^2(z - T) & z > T \end{cases}$$

where N_0 is the background buoyancy frequency and T is a parameter corresponding to the initial peak of the profile.

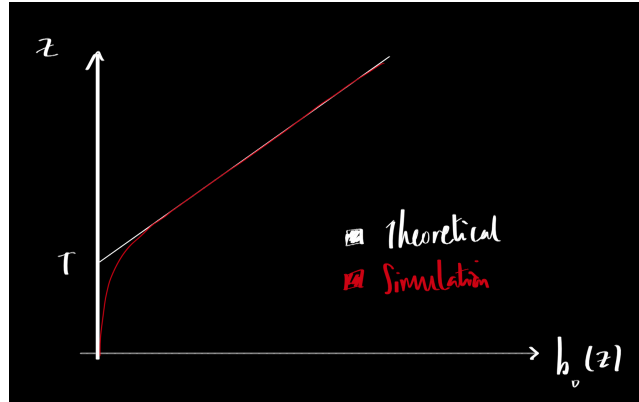


Figure 1: Initial Buoyancy Profile

Note that in simulations, to ensure a smooth buoyancy gradient, we approximate the initial profile with the "softplus" profile defined by

$$b_{\text{softplus}}(z) = \frac{N_0^2}{\text{sharp}} \log(1 + \exp(\text{sharp} * (z - T)))$$

as shown in the above diagram.

After Mixing

After mixing, we expect the following profile:

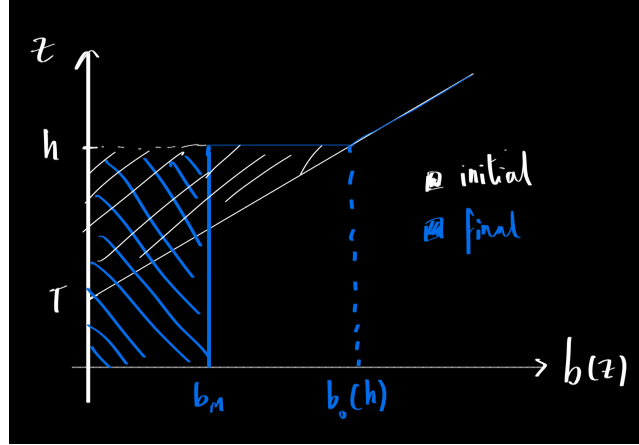


Figure 2: Change in the buoyancy profile

Call the final buoyancy profile $b_f(z)$, where $b_f(z)$ is defined by

$$b_f(z) = \begin{cases} b_m & z < h \\ b_0(z) & z > h \end{cases}$$

Next we can use conservation of buoyancy to find b_m , the buoyancy at the mixed layer height.

$$\int_0^h b_f(z) - b_0(z) dz = 0$$

$$b_m h = \frac{h - T}{2} * b_0(h) = \frac{N_0^2}{2} (h - T)^2$$

Here we make the assumption that the mixed layer height h is greater than T . Rearranging

$$b_m = \frac{N_0^2 (h - T)^2}{2h}$$

From this we can calculate the change in buoyancy

$$\Delta b = b_0(h) - b_m = N_0^2 \left(h - T - \frac{(h - T)^2}{2h} \right) = \frac{N_0^2 (h^2 - T^2)}{2h}$$

Change in potential energy

Next we look at the potential energy. Where necessary, use a control volume with height H such that H is sufficiently larger than the mixed layer height. We

use the relation

$$E_p = - \int z * b(z) dz$$

Looking at the box from 0 to h , initially we have

$$E_p^{\text{initial}} = - \int_0^h z * b_0(z) dz = - \int_T^h z * N_0^2(z - T) dz = -N_0^2 \left(\frac{1}{3} h^3 - \frac{1}{2} T h^2 + \frac{1}{6} T^3 \right)$$

Then for the final profile

$$E_p^{\text{final}} = - \int_0^h z * b_m dz = - \frac{1}{2} h^2 b_m = - \frac{N_0^2 h (h - T)^2}{4}$$

The change in potential energy becomes

$$\begin{aligned} \Delta E_p &= N_0^2 \left(-\frac{h(h - T)^2}{4} + \frac{1}{3} h^3 - \frac{1}{2} T h^2 + \frac{1}{6} T^3 \right) \\ &= N_0^2 \left(\frac{1}{12} h^3 + \frac{1}{6} T^3 - \frac{1}{4} T^2 h \right) \end{aligned}$$

Taking derivatives with respect to h

$$\frac{dE_p}{dh} = N_0^2 \left(\frac{1}{4} h^2 - \frac{1}{4} T^2 \right) = \frac{N_0^2 (h^2 - T^2)}{4} = \frac{h \Delta b}{2}$$

which can very conveniently be written in terms of Δb .

Time differentiation

Taking the time derivative of E_p and applying chain rule, we get

$$\frac{dE_p}{dt} = \frac{dE_p}{dh} * \frac{dh}{dt} = \frac{h \Delta b}{2} \frac{dh}{dt}$$

I suspect there is a simpler way to arrive here given the nice result.

We use the (plane-averaged) buoyancy flux equation

$$\frac{\partial b}{\partial t} + \frac{\partial F_b}{\partial z} = 0$$

where F_b is the buoyancy flux.

Differentiating the integral definition of E_p with respect to time

$$\frac{dE_p}{dt} = - \int_0^H z * \frac{\partial b}{\partial t} dz = \int_0^H z * \frac{\partial F_b}{\partial z} dz$$

We can integrate by parts, noting flux at $z = 0$ and $z = H$ is zero (as H is taken to be large) to get

$$\frac{dE_p}{dt} = - \int_0^H F_b dz$$

Then we use the relation

$$F_b = -K_T \frac{\partial b}{\partial z}$$

to get

$$\frac{dE_p}{dt} = \int_0^H K_T \frac{\partial b}{\partial z} dz = K_T(h) \Delta b$$

We can pull K_T out as above the interface $K_T = 0$ and inside the mixed layer the buoyancy gradient is small/zero. Then equating to the previous expression for $\frac{dE_p}{dt}$ and cancelling the factor of Δb , we end with

$$\frac{dh}{dt} = \frac{2K_T(h)}{h}$$

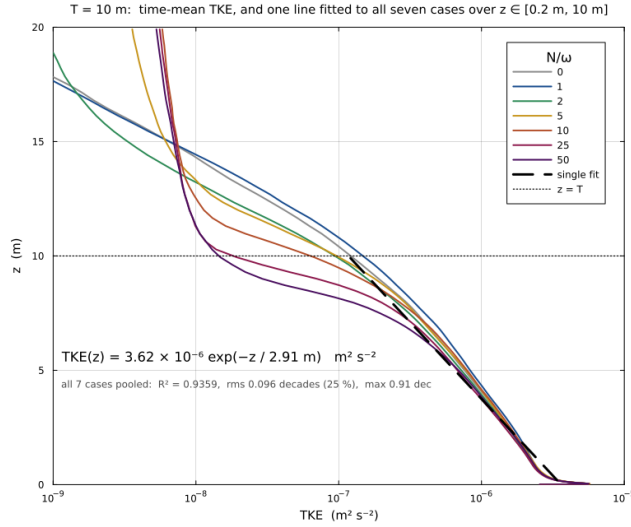
Using simulation data

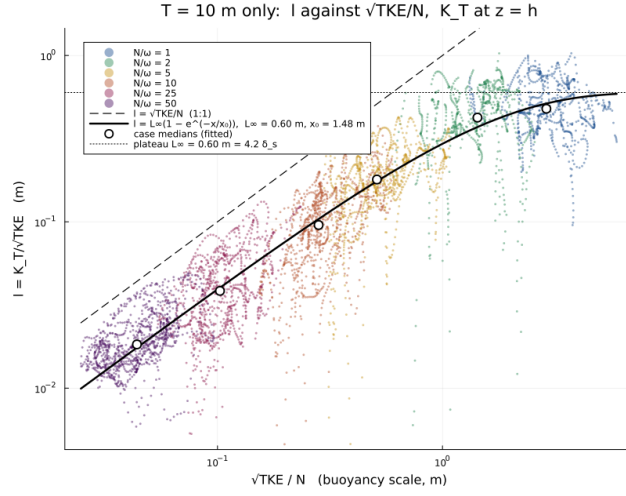
For the Stokes simulations, we found explicit relations between TKE and depth as well as K_T as a function of TKE and buoyancy frequency N .

$$\text{TKE}(z) = 3.62 \times 10^{-6} \exp\left(-\frac{z}{2.91}\right)$$

and

$$K_T(\text{TKE}, N) = 0.6 \sqrt{\text{TKE}} \left(1 - \exp\left(-\frac{\sqrt{\text{TKE}}}{1.48N}\right)\right)$$





Combined with the time evolution relation above, we can numerically find a solution for the growth of the mixed layer height h .

The equation is solved using MATLAB ode45, assuming we start from $h = 10$. For $N = 1$, this gives the following solution

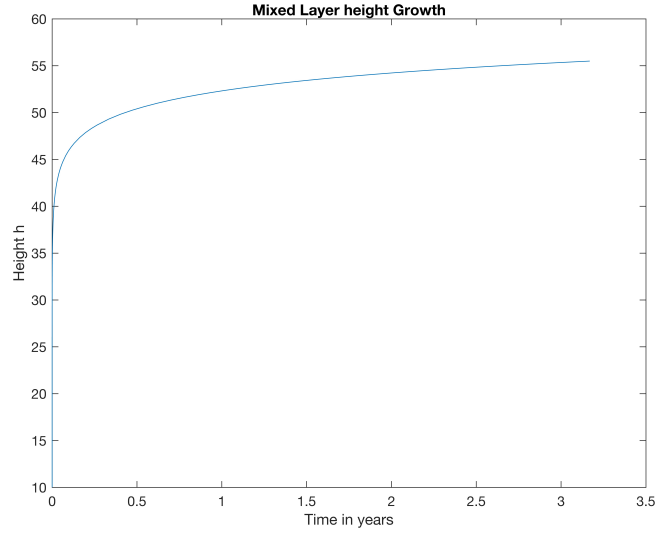


Figure 3: Mixed Layer Height

We can fit this to a log graph

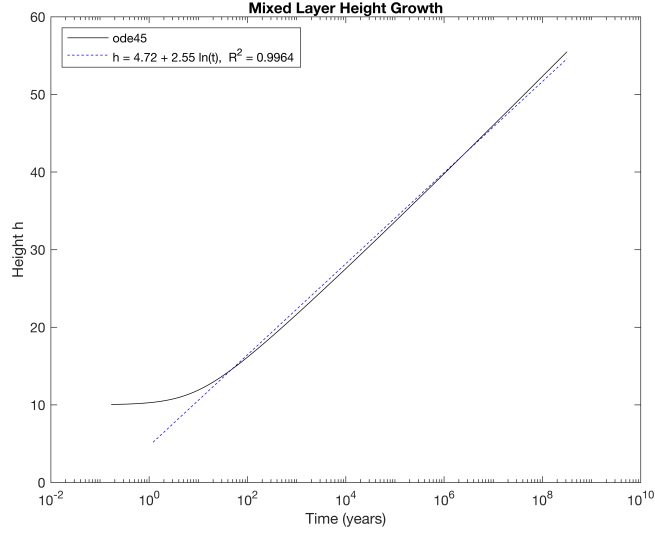


Figure 4: Mixed Layer Height Log Fit

Using the numerical solutions, the estimates for the mixed layer growth times are

h (m)	Elapsed t (yr)	Local τ (yr)
10	0	3.4×10^1
20	5.1×10^2	4.2×10^3
30	2.6×10^4	2.9×10^5
40	1.1×10^6	1.6×10^7
50	4.3×10^7	7.8×10^8

Table 1: Elapsed time and local growth timescale $\tau(h) = h/(dh/dt) = h^2/2K(h)$

Repeating for $N = 0.01$

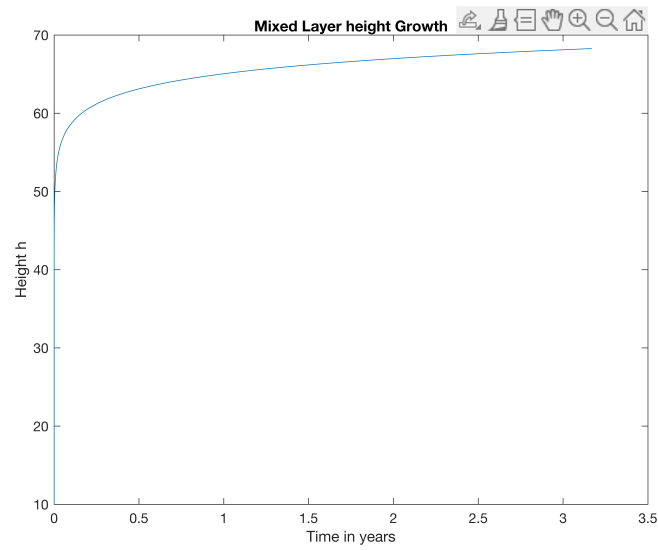


Figure 5: Mixed Layer Height $N = 0.01$

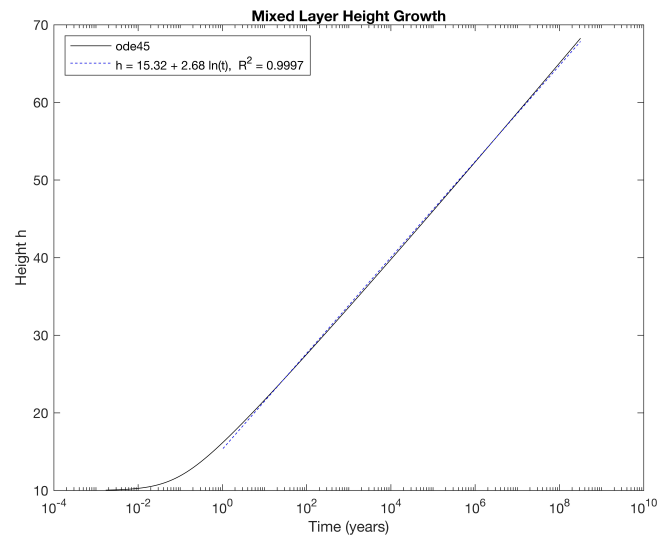


Figure 6: Logfit for $N = 0.01$

h (m)	Elapsed t (yr)	Local τ (yr)
10	0	3.4×10^{-1}
20	5.1	4.2×10^1
30	2.6×10^2	2.9×10^3
40	1.1×10^4	1.6×10^5
50	4.3×10^5	7.8×10^6

Table 2: $N = 0.01$